# RUIZE RICHARD XU

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## EDUCATION

**The University of Chicago**,

*Ph.D. in Computer Science, (GPA:4.0/4.0) advisor: Zhiyuan Li and David McAllester* — Chicago, IL, Expected June 2028
*Bachelor of Science in Computer Science, Mathematics, and Economics*, (GPA: 3.72/4.0) — Chicago, IL, June 2023
Honors Included: Dean's List, ACM-ICPC 2022 Mid-Central USA Gold Medal, Advanced Scholar, Jeff Metcalf Fellowship Grant

## INDUSTRY EXPERIENCE

**Google,** Software Engineer, — Mountain View, CA, June 2024 - September 2024

- Implemented Gmail Ads auction Eligibility Filter and optimized the auction eligibility filter with Reserve Price Optimization
- Experimented with Google DeepMind's FunSearch, an LLM powered code generation algorithm, to optimize the Eligibility Filter to achieve 3% revenue gain in experimental traffic

**Google,** Software Engineer, — Sunnyvale, CA, June 2023 - September 2023

- Implemented Vectorized Order-Preserving Lookup Join algorithm into the Google F1 query kernel execution team, minimizing the cost of GoogleSQL query execution with expensive sort operators and data sharding
- Contributed to existing Google F1 query tests that specifically assess query performance in ordered inputs for table join algorithms

## RESEARCH PROJECT EXPERIENCE

**Robot Intelligence through Perception Lab,** Research Assistant, advisor: Matthew Walter, — Chicago, IL, July 2026 - Now

- Real-to-sim-to-real: trained image-based policies in simulation with domain randomization for millimeter-precision drawer and charger insertion on a low-cost robot arm, deployed on the real robot (CoRL 2026 Late-Breaking Results)

**MathAI and Large Language Models** Research Assistant, advisor: David McAllester, Zhiyuan Li — Chicago, IL, January 2024 - Now

- Developed AO-star based tree search algorithm and reinforcement learning algorithm (variant of GRPO) for Automated Theorem Proving in Lean4 and LLM mathematics reasoning.
- Optimized server hosting LEAN for code sandbox execution for distribtued training.
- Built Leanify, verified proof compression and reusable lemma discovery for Lean with kernel-checked correctness.
- Trained a small transformer to execute unseen programs in MicroPy with PENCIL scaffolding (Findings of EMNLP 2026).

**Higher Order Graph Neural Network,** Research Assistant, advisor: Risi Kondor — Chicago, IL, September 2023 - Now

- Developed Subgraph Level Message Passing Neural Network with PyTorch Geometric and Ptensors (custom C++ package written in Cuda, jointly developed with PhD advisor Risi Kondor), with model achieving State-of-the-art results in chemical property prediction in the ZINC, MolHIV datasets.
- Explored application of Gated Message Passing Graph Neural Networks on Combinatorial Optimization problems with Reinforcement Learning

**System and Machine Learning group,** Research Assistant, advisor: Yanjing Li — Chicago, IL, January 2022 - June 2023

- Experimented binary neural networks (BNN) and different memory and cache design's impacts on the overall inference time with RISC-V on gem5
- Optimized memory and cache-specific BNN by exploiting vector processor parallelism, which achieves 90% speedup for model inference.

## PUBLICATIONS

**Real-to-Sim-to-Real at the Last Millimeter** — CoRL 2026 Late-Breaking Results

Tianchong Jiang*, ***Ruize Xu****, Yicheng Li, Xiangshan Tan, Matthew Walter

**Towards Training Transformers as a Universal Computer** — Findings of EMNLP 2026

***Ruize Xu***, Chenxiao Yang, Yanhong Li, David McAllester

**Teaching an Agent to Sketch One Part at a Time** — ECCV 2026

Xiaodan Du*, ***Ruize Xu****, David Yunis*, Yael Vinker, Greg Shakhnarovich

**Schur Nets: Exploiting Local Structure for Equivariance in Higher Order Graph Neural Networks** — NeurIPS 2024

Qingqi Zhang*, ***Ruize Xu****, Risi Kondor

**Leanify: Verified Proof Compression and Reusable Lemma Discovery** — In Submission

***Ruize Xu***, Anxin Guo*, Yi Gu*, Tianchong Jiang*, Chenxiao Yang, Tianhao Wang, Zhiyuan Li

** Equal contribution*

## SKILLS

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|---|---|
| **Coding Languages** | Python, C, Rust, C++, Haskell, Standard ML, SQL, MATLAB, R, PHP, LaTeX |
| **Frameworks** | PyTorch, Ray, VeRL, PyTorch Geometric, Deep Graph Library (DGL), OpenAI, OpenCV, Unity, NumPy, Pandas, MapReduce |
| **Tools** | vLLM, SGLang, Git, SVN, SSH, SCP, Docker, Slurm, Wireshark, Postgres, SQLite, Vim |
| **Soft Skills** | Language Modeling, Reinforcement Learning, Approximation Algorithm, Graph Theory, Functional Programming |
| **Languages** | Chinese, English |